# SinMULU-JR at the Sinhala MMLU Shared Task (ICTer 2026): English Reasoning over Sinhala Questions with Gemma-3-4B

Technical report: data, method, every experiment, and the first submission

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**Abstract**

We describe our system and the complete experimental record for the Sinhala MMLU Shared Task at ICTer 2026, whose test set consists of 2,238 four-option multiple-choice questions from Sri Lankan 2025 school examinations (Easy, Medium, Hard). Under the task's 8B-parameter, open-weight, no-retrieval rules and a time budget of a few days, we use a single model, `google/gemma-3-4b-it` (4.30B parameters), without fine-tuning, and study 40 inference-time conditions: prompt wording, subject and role information, few-shot examples, option-order averaging, contextual calibration, cloze scoring, 4-bit quantisation, a pretrained base model, chain-of-thought reasoning in Sinhala and in English, and three Sinhala→English translation pipelines (NLLB-200, MADLAD-400, self-translation). Conditions were screened on a 280-question stratified subset of the public (Easy-only) training data and the four best were confirmed on 1,483 held-out questions under a selection rule fixed before the confirmation results were seen. **Letting Gemma reason step by step in English about the Sinhala question, then scoring the answer digit, raises accuracy from 38.2% to 45.6% (+7.4 points, Holm-corrected $p = 1.2 \times 10^{-6}$, $n = 1{,}483$).** Translation pipelines, calibration, option-order averaging, and cloze scoring did not help. We also report a scoring pitfall that silently reduces next-token evaluation of Gemma to chance: the space before the answer digit is a separate token, so the reply must be pre-filled with "පිළිතුර: " (with a trailing space); without it only 5% of the probability falls on the four digits. The first submission (English reasoning with a teacher role) has been produced and validated; its leaderboard score is [to be filled in after the Codabench upload].

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# 1 Introduction

SinhalaMMLU [1] is the first multitask language-understanding benchmark for Sinhala, built from Sri Lankan national school curricula and examinations, with questions grouped by difficulty: Easy (grades 6–8), Medium (O/L level) and Hard (A/L level). The paper finds that open models up to 8B parameters are close to random (25%) and that even Qwen2.5-7B-chat reaches only 30.85% on Easy questions zero-shot (Table 15 of the paper), while Claude 3.5 Sonnet reaches 67.65% overall. The ICTer 2026 shared task asks participants to improve on this under strict constraints (Section 2).

Our constraints were mainly time and compute: work on the final system started on 24 October, three days before the leaderboard closed, on a single rented RTX 3090. We therefore chose one strong small multilingual model and investigated only inference-time methods, with a statistically careful protocol so that the many comparisons would not mislead us.

**Contributions.**

1. A reproducible evaluation pipeline for Sinhala MCQs (`sinmmlu/`) with label-mass validity checks, paired significance tests, and a strict separation of screening, confirmation and test data.
2. A systematic screen of 40 inference-time conditions for Gemma-3-4B-it on Sinhala exam questions, and a pre-registered confirmation of the best four on 1,483 held-out questions.
3. The finding that *English* chain-of-thought over the untranslated Sinhala question gives a robust gain of about +7 points, while explicit machine translation (NLLB, self-translation) does not help.
4. A documented scoring pitfall (answer pre-fill and the space token) and several other engineering errors that we found and corrected, with their effect on results.

# 2 The shared task

## 2.1 Rules

| Rule | How our system complies |
|---|---|
| $\leq$ 8B total parameters across all inference components (MoE counts all experts, LoRA adapters add to the base, quantisation does not reduce the count) | One model: Gemma-3-4B-it, 4,300,079,472 parameters including its unused vision tower (Section 4.2). |
| Open-weight models released before 1 October 2026 | Gemma 3 was released in March 2025 (Gemma licence). |
| No retrieval of any kind (RAG, BM25, dictionaries, lookup tables) | None. Few-shot examples are *fixed per subject* and never chosen by similarity to the test question. |
| Closed APIs only for generating training data, with disclosure | No closed API was used at all. |
| No training, tuning or prompt design on test questions | Every choice was made on the public training data; test questions were only scored to write predictions (one process note in Section 11). |
| Organisers reproduce every final system from submitted code | Deterministic greedy decoding, pinned library versions, model revision hash, one command per step (Section 10). |
| Cite Pramodya et al., EMNLP 2025 | [1]. |

**Table 1:** Task rules and compliance.

## 2.2 Metric, submission format and schedule

The leaderboard metric is the **macro average of the Easy, Medium and Hard accuracies**. Submissions are made on Codabench (at most 3 per day) as a zip containing `predictions.jsonl`, one line per test question, e.g. `{"q_no": 1, "subject": "Art", "category": "Humanities", "level": "easy", "answer": "2"}`. The `answer` is a string; the key fields must be copied verbatim from the test files (their subject strings are inconsistent, Section 3). Schedule: leaderboard closes 27 October 2026 (23:59), code due 28 October, system report and slides due 30 October.

# 3 Data

## 3.1 Public training and test sets

| Set | Questions | Files (subjects) | Options | Years / level |
|---|---|---|---|---|
| Public train (with answers) | 1,851 | 14 | 4 | 2018–2023, Easy only (grades 6–8) |
| Public test, Easy | 642 | 13 | 4 | 2025 |
| Public test, Medium | 1,111 | 13 | 4 | 2025 |
| Public test, Hard | 485 | 15 | 4 | 2025 |
| Public test, total | 2,238 | 41 | 4 | no answers |

**Table 2:** Public data provided by the organisers.

**Training-set subjects.** Arts 100, Buddhism 162, Catholicism 132, Christianity 130, Civics 123, Eastern music 143, Geography 117, Health and Physical Science 130, History 151, Islam 108, Sinhala language and literature 155, Dancing 108, Drama and theatre 128, Science 164. Answer distribution: 1: 473, 2: 501, 3: 507, 4: 370 (option 4 is under-represented, 20.0%).

**Test-set subjects.** Easy: the same 13 school subjects as above minus Islam (Art 30, Buddhism 30, Catholicism 50, Christianity 58, Civics 30, Dancing 39, Drama 60, Eastern Music 44, Geography 50, Health 36, History 40, Science 92, Sinhala 83). Medium (O/L): Agriculture and Food Technology 37, Buddhism 174, Catholicism 40, Christianity 38, Civics 159, Communication and Media Studies 40, Dancing 153, Design and Construction Technology 37, Eastern Music 36, Geography 41, Health 72, History 125, Sinhala 159. Hard (A/L): Agricultural Science 37, Art 15, Biology 50, Biosystem Technology 41, Buddhist Civilization 40, Business Studies 29, Chemistry 14, Communication and Media Studies 50, Dancing (Indigenous) 22, Economics 46, Geography 40, Home Economics 29, Physics 22, Political Science 30, Sinhala 20. Many Medium and Hard subjects (e.g. Biology, Economics, Political Science) have no training data at all.

**Data issues found and handled.**

- **Label error:** Geography (training file `Geography_social_science_easy_f_117.json`) question 64 had an invalid answer; it was corrected to 3 (2026-10-14).
- **Inconsistent subject strings** inside one file, e.g. "Eastern_Music" and "Eastern Music", "drama and Theatre " (trailing space), and different English names between train and test ("Arts"/"Art", "Health and Physical Science"/"Health"). Submission keys are copied verbatim; internally every question gets a unique id `<level>/<file>/q<q_no>`, and the source file is used as the subject key.
- **Zero-width characters:** the zero-width non-joiner (U+200C) appears in 357 of 2,238 test items but only 29 of 1,851 training items (a legacy-font artefact); it is removed from model inputs. The zero-width joiner (U+200D) is kept because it is part of Sinhala conjuncts such as ශ්‍රී.

**Data we did not use.** The Hugging Face release `naist-nlp/SinhalaMMLU` was *deliberately not accessed*: its card states that the shared-task Hard test set ("SinhalaMMLU 2.0") is built from the paper's Hard questions with one option removed, so it may contain test questions with answers. It is also gated under CC BY-NC-ND 4.0. No other external data was used for the system described here.

## 3.2 Contamination check

Every training question was compared with every test question by character 5-gram Jaccard similarity on a normalised text (NFC, ZWJ/ZWNJ removed, punctuation and symbols mapped to spaces, lower-cased Latin, blank-runs such as "…" and "_____" removed), computed two ways and exactly for every candidate pair: *full* = question + options sorted alphabetically (years reorder options), and $q$ = question only (only for questions of $\geq 25$ characters). Classes: remove if full $\geq 0.5$; review if full is in $[0.35, 0.5)$ or ($q \geq 0.5$ and full $\geq 0.2$); ignored if $q \geq 0.5$ and full $< 0.2$ (generic stems such as "choose the correctly spelled word"). Stricter thresholds (0.35/0.25) are defined for collected past papers.

| Candidates vs. public test (2,238) | remove | review | ignored |
|---|---|---|---|
| Public train (1,851) | 18 pairs / 18 items | 100 pairs | 21 pairs |

**Table 3:** Near-duplicates between public train and test. The 18 remove items (full Jaccard 0.5–0.83) are excluded from every evaluation set used for decisions (`easy_clean`).

### 3.3 Evaluation sets

Because the team decided to keep the public training data for *evaluation only*, all model selection uses it as a development set. All sets are built deterministically by `python -m sinmmlu.build_dev` (seed 42); their id lists are version-controlled, and rebuilding on the pod gave byte-identical id lists (MD5 checked).

| File | $n$ | Content and use |
|---|---|---|
| `easy_full` | 1,851 | all public train |
| `easy_clean` | 1,833 | minus the 18 test near-duplicates |
| `inference_subset` | 280 | **screening set**: 20 random questions from each of the 14 subjects of `easy_clean` (seed 42). Gold answers 1/2/3/4 = 77/74/74/55, so always answering "1" gives 27.5% (the chance line); 95% interval about ±5.6 points. |
| `fewshot_pool` | 70 | 5 fixed examples per subject, not in the subset (seed 43). The first four of each subject have four different answers, so that a $k$-shot prompt does not teach a position bias. The only source of few-shot examples. |
| `inference_rest` | 1,483 | **confirmation set**: the rest of `easy_clean`; disjoint from the two above; 95% interval about ±2.5 points. |

**Table 4:** Evaluation sets (all from public train, all Easy).

## 4 Model choice

### 4.1 Phase 0: small models

Before the final week, the team screened three small instruction-tuned models on all 1,851 Easy questions with next-token scoring of the digits 1–4 (Table 5). All were at chance. Tokeniser *fertility* (tokens per whitespace-separated Sinhala word, over all 1,851 questions) strongly favoured Gemma: 2.96 for Gemma-3-1B-it, against 8.66 for Qwen3-0.6B and 10.74 for Llama-3.2-1B-Instruct. Gemma's prompts are therefore about three times shorter, and its vocabulary covers Sinhala far better.

| Model (chat template on) | 1 order | 4 orders | calibrated | label mass |
|---|---|---|---|---|
| Qwen3-0.6B ("strict" prompt) | 25.6 | 24.9 | 26.6 | 0.010 |
| Gemma-3-1B-it | 26.6 | 28.3 | 26.7 | 0.984 |
| Llama-3.2-1B-Instruct | 25.6 | 23.7 | 26.5 | 0.873 |

**Table 5:** Phase 0 (14–15 October, Colab T4, full precision), accuracy (%) on `easy_full` ($n$=1,851, ±2.0 points). "Label mass" is the total probability on the four digit tokens. With the chat template off, every model had a label mass of 0.000; Section 5.2 explains why.

### 4.2 Gemma-3-4B-it

Given the time available, we chose `google/gemma-3-4b-it`[1] (context length 131,072 tokens), loaded in bfloat16. It is a multimodal checkpoint; we use text input only, but count all its parameters. Parameters counted from the stored safetensors: **4,300,079,472** = 3,880,263,168 (language model) + 419,816,304 (vision tower and projector).

[1] Hugging Face revision `093f9f388b31de276ce2de164bdc2081324b9767`.

**Correction.** Our first parameter counter summed the parameters of an un-initialised (meta-device) model without tying shared matrices and therefore counted tied embeddings twice: it reported 4.97B for Gemma-3-4B, 3.46B for MADLAD-400-3B and 1.40B for NLLB-600M. The correct values are 4.30B, 2.94B (MADLAD stores an untied output layer) and 0.62B. An earlier statement in our notes that Gemma + MADLAD ("8.44B") exceeds the limit was therefore wrong: the combination has 7.24B parameters and is allowed. This did not affect any decision (Section 7.2).

# 5 Method

## 5.1 Label scoring

For a question $x$ with options $o_1, \ldots, o_4$, a prompt $\pi(x)$ is rendered (with Gemma's chat template unless stated otherwise), and the model's next-token distribution is read at the answer position. For each label $\ell \in \{1, 2, 3, 4\}$ we take $\log P(\ell \mid \pi(x))$ over the *full* vocabulary, renormalise over the four labels, and predict the argmax. This is the scoring used by the SinhalaMMLU paper for open models. We record the **label mass** $\sum_\ell P(\ell \mid \pi(x))$ for every question: if it is low, the model wanted to say something else, and the argmax over the four labels is weak evidence. A condition is only trusted if its mean label mass is $\geq 0.5$. Prompts are never truncated (an over-long prompt raises an error), batches are left-padded with explicit position ids, and any non-finite logit stops the run.

## 5.2 The answer pre-fill and the space token

The first test run of the pipeline on Gemma-3-4B-it showed a mean label mass of only about 0.05. Inspecting the top next tokens revealed two causes. (i) Gemma begins its reply with a phrase such as "වැඩිම ගැළපෙන පිළිතුර: 2 …" ("the most suitable answer: 2") or with "**" (Markdown bold), so the digit is not the first token. (ii) Gemma's tokenizer encodes "␣2" as two tokens, a space and "2", and has no single token for the digit with a leading space, so after "…:" the model's next token is the space. Table 6 shows the effect on 40 questions.

| Rendering | Text appended after the prompt | Mean label mass |
|---|---|---|
| chat template | nothing (generation prompt only) | 0.053 |
| chat template | "පිළිතුර:" | 0.000 |
| chat template | **"පිළිතුර:␣"** | **1.000** |
| chat template | "පිළිතුර:␣**" | 1.000 |
| raw text (no template) | nothing (prompt ends with "පිළිතුර:") | 0.001 |
| raw text | **"␣"** | **0.999** |
| chat, English instructions | **"Answer:␣"** | **1.000** |

**Table 6:** Answer pre-fill probe, Gemma-3-4B-it, first 40 questions of the subset. ␣ marks the trailing space.

All label-scored prompts are therefore scored with the reply pre-filled: in chat mode the model turn starts with "*answer-word*:␣" ("පිළිතුර" for Sinhala prompts, "Answer" for English ones), few-shot example answers are written the same way, and raw prompts get a trailing space. Only the bare-digit token is scored. With this, the label mass is $\geq 0.997$ for every valid condition. The zero label mass of all chat-off runs in phase 0 (Table 5) is consistent with the same space-token effect. We did not test whether it affects other published next-token evaluations of Sinhala.

## 5.3 Option-order averaging and contextual calibration

**Averaging** scores each question under the 4 cyclic option orders $\sigma_s(i) = (i + s) \bmod 4$, maps the four renormalised label probabilities back to the original options, and averages them. **Contextual calibration** [2] estimates the model's label prior from three content-free inputs ("N/A", the empty string, and "[MASK]", each used as the question and as all four options), averages their log-probabilities, and subtracts this prior before renormalising. When the prompt shows the subject, one prior is estimated per subject.

## 5.4 Cloze scoring

The question is shown *without* options, and each option text $o_i$ is scored as a continuation of the pre-filled answer: $\frac{1}{|o_i|} \sum_t \log P(o_{i,t} \mid \pi(x), o_{i,<t})$ (mean log-probability per token). The argmax over options is the

prediction.

### 5.5 Reasoning (chain of thought)

The model first generates a reasoning text with greedy decoding (at most 512 new tokens in English, 768 in Sinhala, since Sinhala needs about three times as many tokens per word). We then (i) find the last "Answer: *d*" or "පිළිතුර: *d*" it wrote (the *parse rate* is the fraction of questions where one is found), (ii) cut the reasoning just before it, (iii) append the cue "\n\n Answer:␣" (or "පිළිතුර:␣"), and (iv) score the four labels at that position as above. Every question therefore gets a probability vector, also when the reasoning was cut off by the length limit; where an answer was stated, the scored answer agrees with it in 99.6–100% of cases. Generation is deterministic (greedy, fixed seed 42), and the reasoning texts are cached so that an interrupted run resumes.

### 5.6 Few-shot examples

$k$-shot prompts ($k \in \{1, 3, 5\}$) put $k$ solved examples *of the same subject* before the question, as earlier chat turns (user: the example question; assistant: "පිළිතුර: *d*"). The examples come only from the fixed `fewshot_pool`, in a fixed order, so every question of a subject sees the same examples; nothing is chosen by similarity to the question. Test subjects are matched to training subjects by the first three letters of the English subject name ("Art"↔"Arts", "Health"↔"Health and Physical Science", "Sinhala"↔"Sinhala language and literature"); subjects without training examples (most Medium and Hard subjects) use a fixed generic set (the first example of each subject).

**Worked examples for reasoning.** For `cot_en_fewshot2`, Gemma generated English reasoning (prompt `cot_en`) for the 70 pool questions; the 28 whose stated answer was correct were kept, up to and including their "Answer: *d*" line, as two worked examples per subject. Buddhism had no correct rationale and uses the generic fallback. These rationales are model-written, from public training data only.

### 5.7 Translation pipelines

The question and each option are translated separately into English and cached:

- **NLLB**: `facebook/nllb-200-distilled-600M` [3] (revision `f8d333a0…`, 615M parameters, CC-BY-NC 4.0), source `sin_Sinh`, target `eng_Latn`.
- **MADLAD**: `google/madlad400-3b-mt` [4] (revision `fa184c67…`, 2.94B parameters, Apache 2.0), input prefix "`<2en>`".
- **Self**: Gemma-3-4B-it itself, prompted "Translate the following Sinhala text into English. Write only the English translation, nothing else." (greedy, $\leq 512$ new tokens).

NLLB and MADLAD are sentence-level systems: texts are split into lines and sentences (after ".", "?", "!"), each piece is translated with beam search (4 beams, $\leq 256$ new tokens), and the pieces are re-joined; pieces without Sinhala letters are copied unchanged. Gemma then answers one of three prompt variants: English only, bilingual (Sinhala original followed by the English translation, with a note that the translation may contain errors), or the Sinhala question with English options. On the subset, NLLB translated 1,402 unique sentence pieces in about a minute; MADLAD translated 1,437 pieces (several minutes, run time not recorded).

### 5.8 Ensembles

An ensemble averages the four option probabilities of finished conditions of the *same* model (so the parameter count does not grow) and takes the argmax.

### 5.9 Prompts

All prompts are in Appendix A. The baseline `paper_nosubject` is Appendix B of the paper without the subject line.

## 6 Experimental protocol

**Stages.**

1. **Smoke test**: 8 questions through every code path (no results recorded). It found the pre-fill problem.

2. **Screening** on `inference_subset` ($n$=280) in three waves: (1) prompt wording, chat template, averaging, calibration, cloze, 4-bit, model references; (2) few-shot, NLLB and self-translation; (3) reasoning in Sinhala and English, MADLAD.
3. **Combinations** of the screening winners (also on the subset).
4. **Confirmation** on `inference_rest` ($n$=1,483) under a rule written into our lab notebook before the combination results were seen: take the four best *valid* conditions on the subset (label mass $\geq 0.5$, parameter budget respected, ensembles only within one model) plus the baseline; a condition is *adopted* if it beats the baseline by $\geq 1$ point with Holm-corrected exact McNemar $p < 0.05$ over the confirmed comparisons; the most accurate adopted condition wins; if none is adopted, the baseline is submitted.
5. **Test**: the winner is run once on the 2,238 test questions.

**Statistics.** Accuracy with a 95% Wilson interval; for every condition against the baseline on the same questions: the exact two-sided McNemar test on the discordant pairs, Holm–Bonferroni correction over all conditions compared in that table, and a paired bootstrap 95% interval of the accuracy gain (10,000 resamples of questions, seed 0).

**Hardware and software.** RunPod pod with one NVIDIA GeForce RTX 3090 (24 GB), 128 vCPUs, 251 GB RAM; Python 3.12.3, PyTorch 2.8.0 (CUDA 12.8), Transformers 5.18.0, Accelerate 1.15.0, bitsandbytes 0.50.2, PEFT 0.21.1. Gemma in bfloat16 with batch size 16 for scoring and generation; batches are split in halves automatically on GPU out-of-memory. 4-bit runs use NF4 with bfloat16 compute.

# 7 Results

## 7.1 Screening (280 questions)

Table 7 lists every condition. The baseline reaches 37.5%, already well above the 27.5% chance line.

**Table 7:** Screening on `inference_subset` ($n$=280; chance 27.5%). Gain = accuracy minus baseline (points); CI = paired bootstrap 95% interval of the gain; $p$ = exact McNemar; $p_{\mathrm{H}}$ = Holm-corrected over the 39 comparisons of the final table; L = mean label mass. All with Gemma-3-4B-it, chat template, one option order, no calibration, bf16, unless stated.

| Condition | Acc. | Gain | CI of gain | $p$ | $p_{\mathrm{H}}$ | L |
|---|---|---|---|---|---|---|
| *A: baseline* | | | | | | |
| A0 `paper_nosubject` | 37.5 | – | – | – | – | 1.000 |
| A0b same, chat template off | 37.5 | +0.0 | [−4.3, +4.3] | 1.00 | 1.00 | 0.997 |
| *B: prompt wording* | | | | | | |
| B1 `paper_subject_si` (Sinhala subject name) | 42.1 | +4.6 | [+0.0, +9.6] | 0.092 | 1.00 | 1.000 |
| B2 `paper_subject_en` (English subject name) | 40.7 | +3.2 | [−1.8, +8.2] | 0.262 | 1.00 | 1.000 |
| B3 `strict_nosubject` ("write only the digit") | 36.8 | −0.7 | [−5.0, +3.2] | 0.868 | 1.00 | 0.997 |
| B4 `role_subject_si` (Sinhala teacher role + subject) | 43.6 | +6.1 | [+1.1, +11.1] | 0.030 | 0.87 | 1.000 |
| B5 `en_instr_nosubject` (English instructions) | 41.8 | +4.3 | [−0.4, +8.9] | 0.088 | 1.00 | 1.000 |
| B6 `letters_nosubject` (labels A–D) | 36.4 | invalid: label mass 0.002 | | | | 0.002 |
| B7 `fewshot1_nosubject` | 40.7 | +3.2 | [−1.8, +8.6] | 0.281 | 1.00 | 1.000 |
| B7 `fewshot3_nosubject` | 42.5 | +5.0 | [−0.7, +10.7] | 0.120 | 1.00 | 1.000 |
| B7 `fewshot5_nosubject` | 42.1 | +4.6 | [−1.1, +10.4] | 0.148 | 1.00 | 1.000 |
| *C: scoring and model* | | | | | | |
| C1 4 option orders | 38.6 | +1.1 | [−3.6, +5.7] | 0.766 | 1.00 | 1.000 |
| C2 contextual calibration | 34.3 | −3.2 | [−7.5, +0.7] | 0.163 | 1.00 | 1.000 |
| C3 4 orders + calibration | 38.2 | +0.7 | [−4.3, +5.7] | 0.888 | 1.00 | 1.000 |
| C4 cloze (`cloze_nosubject`) | 31.4 | −6.1 | [−13.6, +1.4] | 0.128 | 1.00 | – |
| C6 `cot_si` (reasoning in Sinhala) | 43.2 | +5.7 | [−0.4, +11.8] | 0.085 | 1.00 | 1.000 |
| C7 `cot_en` (reasoning in English) | **47.9** | **+10.4** | [+4.3, +16.4] | 0.0013 | 0.047 | 1.000 |
| C8 4-bit NF4 | 36.8 | −0.7 | [−6.4, +5.0] | 0.901 | 1.00 | 1.000 |
| C9 Gemma-3-4B-**pt** (base model, raw prompt) | 45.0 | +7.5 | [+0.7, +14.3] | 0.046 | 1.00 | 0.917 |
| C9 Gemma-3-1B-it | 25.7 | −11.8 | [−19.3, −4.3] | 0.0032 | 0.12 | 1.000 |
| *D: translation (Gemma answers; translator in brackets)* | | | | | | |
| D1 English only [self] | 37.9 | +0.4 | [−6.4, +7.1] | 1.00 | 1.00 | 1.000 |
| D2 English only [NLLB] | 37.1 | −0.4 | [−7.5, +6.8] | 1.00 | 1.00 | 0.998 |
| D3 English only [MADLAD] | 43.9 | +6.4 | [−0.7, +13.6] | 0.092 | 1.00 | 1.000 |
| D4 bilingual [self] | 37.5 | +0.0 | [−5.7, +5.7] | 1.00 | 1.00 | 1.000 |
| D4 bilingual [NLLB] | 40.4 | +2.9 | [−2.9, +8.6] | 0.382 | 1.00 | 1.000 |
| D4 bilingual [MADLAD] | 42.9 | +5.4 | [−0.4, +11.1] | 0.082 | 1.00 | 0.998 |

| Condition | Acc. | Gain | CI of gain | $p$ | $p_H$ | L |
|---|---|---|---|---|---|---|
| D5 Sinhala question, English options [self] | 35.0 | −2.5 | [−8.9, +3.9] | 0.510 | 1.00 | 0.997 |
| D5 same [NLLB] | 36.1 | −1.4 | [−8.2, +5.4] | 0.755 | 1.00 | 1.000 |
| D5 same [MADLAD] | 37.9 | +0.4 | [−5.7, +6.4] | 1.00 | 1.00 | 1.000 |
| *E: combinations* | | | | | | |
| `cot_en_fewshot2` (English reasoning + 2 worked examples) | **50.7** | **+13.2** | [+7.1, +19.3] | 7.4e-5 | 0.003 | 0.996 |
| `cot_en_role` (English reasoning + teacher role) | 48.2 | +10.7 | [+3.9, +17.5] | 0.0032 | 0.12 | 1.000 |
| `cot_en_subject` (English reasoning + subject) | 46.8 | +9.3 | [+2.9, +15.7] | 0.0073 | 0.24 | 1.000 |
| `fewshot3_subject_si` | 42.9 | +5.4 | [−0.4, +11.1] | 0.096 | 1.00 | 1.000 |
| pt + `fewshot3_nosubject` | 47.1 | +9.6 | [+3.6, +15.7] | 0.0036 | 0.12 | 0.999 |
| pt + `fewshot3_subject_si` | 46.8 | +9.3 | [+3.2, +15.4] | 0.0049 | 0.16 | 0.999 |
| pt + `paper_subject_si` | 40.4 | +2.9 | [−3.6, +9.6] | 0.470 | 1.00 | 0.918 |
| pt + `role_subject_si` | 42.1 | +4.6 | [−2.1, +11.4] | 0.218 | 1.00 | 0.894 |
| ensemble `cot_en` + `cot_si` | 49.3 | +11.8 | [+6.1, +17.9] | 1.4e-4 | 0.005 | – |
| ensemble `cot_en` + B4 + B7(3) | 44.6 | +7.1 | [+2.1, +12.1] | 0.0078 | 0.24 | – |
| ensemble B4 + B7(3) + B1 + B5 | 44.3 | +6.8 | [+2.1, +11.4] | 0.0079 | 0.24 | – |

**Observations on the screening set.**

- **Reasoning in English is the strongest single change** (+10.4, the only single condition significant after Holm). Reasoning in Sinhala helps less (+5.7) and is slower (450 s against 350 s for 280 questions).
- **Telling the model the subject** helps (+3 to +6), as the paper found; a Sinhala teacher role helps most. English instructions alone help about as much (+4.3).
- **Few-shot** examples help modestly (+3 to +5); 3 examples are as good as 5.
- **Option-order averaging** changes little (+1.1): Gemma has no strong position preference after the pre-fill fix. **Calibration hurts** (−3.2): it shifts predictions towards option 2 (46.8% of predictions, against 26.4% of gold answers), i.e. the content-free prior is a poor estimate of the model's bias here.
- **Cloze scoring is worse** (−6.1): Sinhala options of very different lengths and forms are not comparable by mean token log-probability.
- **4-bit NF4 costs nothing measurable** (−0.7, $p$=0.90), relevant if the organisers reproduce on a smaller GPU.
- **The pretrained base model** scores surprisingly well (45.0%) but has a strong preference for option 1 (46.4% of its predictions; 57.5% with the subject prompt) and a lower label mass (0.92), and it cannot follow the chat-style reasoning prompts that won.
- **Letters A–D** are not usable: Gemma answers with a Sinhala letter (e.g. ඊ) or a digit, so only 0.2% of the probability falls on the Latin letters.
- **NLLB and self-translation do not help** (between −2.5 and +2.9). MADLAD translation helps more (+6.4 English only, +5.4 bilingual) but stays below English reasoning; Gemma + MADLAD (7.24B) would be within the parameter limit.
- **Gemma-3-1B-it is at chance** (25.7%).

## 7.2 Selection

By the rule of Section 6, the confirmation candidates were the four best valid conditions on the subset: `cot_en_fewshot2` (50.7), the ensemble `cot_en`+`cot_si` (49.3), `cot_en_role` (48.2) and `cot_en` (47.9). At selection time MADLAD conditions were excluded as over budget because of the counting error of Section 4.2; with the correct count they would have been eligible, but the best MADLAD condition (43.9) ranked twelfth and would not have been selected.

## 7.3 Confirmation (1,483 held-out questions)

All four candidates beat the baseline clearly; the three English-reasoning variants lie within 0.8 points of each other, which is far inside their uncertainty, so the rule's tie-break (highest accuracy) chose `cot_en_role`. Two subset results did not replicate: the worked examples fell from 50.7% to 44.8%, and Sinhala reasoning from 43.2% (+5.7) to 36.8% (−1.3). This is the expected regression of the top of a 280-question ranking, and it is why decisions were taken only on the confirmation set.

**Discordant pairs.** Against the baseline, `cot_en_role` is right on 283 questions where the baseline is wrong and wrong on 173 where the baseline is right: reasoning changes many answers in both directions, with a clear net gain.

| Condition | Acc. | Gain | CI of gain | $p$ | $p_{\mathrm{H}}$ | Adopted | Parse rate | Time (s) |
|---|---|---|---|---|---|---|---|---|
| A0 baseline `paper_nosubject` | 38.2 | – | – | – | – | – | – | 36 |
| `cot_en_role` | **45.6** | **+7.4** | [+4.6, +10.3] | 2.9e-7 | 1.2e-6 | **yes, winner** | 76.6% | 2,084 |
| `cot_en` | 44.9 | +6.7 | [+3.9, +9.6] | 5.5e-6 | 1.7e-5 | yes | 97.0% | 1,871 |
| `cot_en_fewshot2` | 44.8 | +6.6 | [+3.7, +9.6] | 1.6e-5 | 3.2e-5 | yes | 95.5% | 2,791 |
| ensemble `cot_en`+`cot_si` | 44.0 | +5.9 | [+3.2, +8.6] | 3.8e-5 | 3.8e-5 | yes | – | – |
| `cot_si` (ensemble part only) | 36.8 | −1.3 | [−4.0, +1.3] | 0.36 | – | – | 97.4% | 2,400 |

**Table 8:** Confirmation on `inference_rest` ($n$=1,483, ±2.5 points). $p_{\mathrm{H}}$ is Holm-corrected over the four candidates (the selection rule); `cot_si` was run only as a part of the ensemble and is not a candidate. Times are for the 1,483 questions on the RTX 3090.

## 7.4 Per-subject accuracy

| Subject | | $n$ | A0 | `cot_en` | `cot_en_role` | `cot_en_fs2` | `cot_si` |
|---|---|---|---|---|---|---|---|
| Arts | චිත්‍ර | 70 | 42.9 | 44.3 | 45.7 | 42.9 | 50.0 |
| Buddhism | බුද්ධ ධර්මය | 137 | 33.6 | 44.5 | 40.2 | 40.9 | 36.5 |
| Catholicism | කතෝලික ධර්මය | 104 | 33.7 | 37.5 | 42.3 | 40.4 | 33.7 |
| Christianity | ක්‍රිස්තියානි ධර්මය | 103 | 38.8 | 51.5 | 49.5 | 52.4 | 32.0 |
| Civics | පුරවැසි අධ්‍යාපනය | 96 | 57.3 | 61.5 | 62.5 | 63.5 | 53.1 |
| Eastern music | පෙරදිග සංගීතය | 118 | 33.1 | 36.4 | 33.9 | 39.0 | 34.8 |
| Geography | භූගෝල විද්‍යාව | 90 | 34.4 | 52.2 | 40.0 | 50.0 | 41.1 |
| Health and Phys. Sci. | සෞඛ්‍ය හා ශාරීරික විද්‍යාව | 104 | 45.2 | 59.6 | 58.7 | 51.0 | 47.1 |
| History | ඉතිහාසය | 126 | 27.8 | 35.7 | 32.5 | 34.9 | 24.6 |
| Islam | ඉස්ලාම් ධර්මය | 83 | 41.0 | 50.6 | 59.0 | 51.8 | 36.1 |
| Sinhala lang. and lit. | සිංහල භාෂාව සහ සාහිත්‍යය | 130 | 35.4 | 33.1 | 40.0 | 34.6 | 29.2 |
| Dancing | නර්තනය | 82 | 40.2 | 26.8 | 40.2 | 32.9 | 23.2 |
| Drama and theatre | නාට්‍ය හා රංග කලාව | 101 | 35.6 | 46.5 | 37.6 | 42.6 | 38.6 |
| Science | විද්‍යාව | 139 | 42.4 | 51.8 | 60.4 | 54.0 | 41.7 |
| All | | 1,483 | 38.2 | 44.9 | 45.6 | 44.8 | 36.8 |

**Table 9:** Accuracy (%) per subject on the confirmation set (descriptive only; per-subject intervals are ±8–12 points, so no decision was taken per subject).

English reasoning helps most on Geography (+17.8 for `cot_en`), Health (+14.4), Christianity (+12.7), Buddhism (+10.9), Drama (+10.9), Science (+9.4) and Islam (+9.6), i.e. content subjects. It does not help, or hurts, on the most language- and culture-bound subjects: Sinhala language and literature (−2.3) and Dancing (−13.4 for `cot_en`; 0.0 with the teacher role). This matches the paper's observation that culturally specific knowledge is the hardest part of SinhalaMMLU, and suggests that reasoning in English loses information when the question is about the Sinhala language itself.

## 7.5 Comparison with the paper

Our Easy-level numbers are on public-train questions (2018–2023) and the paper's on its own Easy split, so the comparison is indicative only. The paper reports zero-shot Easy accuracy of 30.85% for Qwen2.5-7B-chat, 34.31% for Llama-3.1-8B and 45.11% for Qwen2.5-72B-chat (Table 15 of [1]). Gemma-3-4B-it scored 38.2% with the paper's own prompt (pre-filled), and 45.6% with English reasoning, i.e. at the level the paper reports for a 72B model.

## 7.6 Test predictions (submission 1)

`cot_en_role` was run on all 2,238 test questions in 3,083 s on the RTX 3090 (longest input 961 tokens). The file passed format validation twice (on the pod and locally): 2,238 lines, keys identical to the test set, every answer in {"1","2","3","4"}. The model stated an answer in 62.4% of its reasoning texts (76.6% on the confirmation set), probably because the longer Medium and Hard questions more often reach the 512-token limit; every question still receives a scored answer, and the mean label mass after the cue was 0.998.

| Submission | Easy | Medium | Hard | Macro average |
|---|---|---|---|---|
| 1: `cot_en_role` | [–] | [–] | [–] | [–] |

**Table 10:** Leaderboard results (to be filled in after the Codabench upload).

# 8 Final system

- **Model**: `google/gemma-3-4b-it`, revision `093f9f38…`, bfloat16, 4,300,079,472 parameters (only component; 8B limit respected).
- **Input**: NFC normalisation, ZWNJ removed, ZWJ kept; options numbered "1."–"4."; the English subject name from the test file.
- **Prompt** (user turn of Gemma's chat template):

```
You are an expert Sri Lankan teacher of {subject}.
The following is a multiple choice question in Sinhala about {subject}.
Think step by step in English, then choose the correct or most suitable answer from the options 1, 2, 3, 4.
End your reply with "Answer: <number>".

Question: {Sinhala question}
1. {option 1}
...
4. {option 4}
```

- **Decoding**: greedy, at most 512 new tokens; the stated answer (if any) is cut, "\n\n Answer:␣" appended, and the argmax over the digit tokens 1–4 is the prediction.
- **No** retrieval, no fine-tuning, no external data, no closed API, no test-time adaptation.

# 9 Engineering record: errors found and fixed

Small engineering errors can silently turn a benchmark run into noise. Table 11 lists every error we found, including our own.

**Table 11:** Errors found during the project.

| Problem | Effect | Fix |
|---|---|---|
| YAML 1.1 parses unquoted `chat: on` as the boolean `True` (phase 0) | runs labelled "chat on" were run with the chat template off | chat mode as a command-line choice; old rows quarantined |
| Prompts truncated from the right at 1,024 tokens (phase 0) | the answer position was cut off for 9–78 Qwen/Llama prompts | no truncation; over-long prompts raise |
| Dedup "ignored" class depended on a thresholded zero (phase 0) | generic-stem pairs mislabelled | both Jaccards computed exactly for every pair |
| 4-bit loading used bitsandbytes' default FP4 with fp16 compute; no NaN check | lower 4-bit quality; Gemma 3 can overflow in fp16 | NF4 with bf16/fp32 compute; non-finite logits raise |
| The experiment grid forced full precision | 4-bit runs impossible | `--quant` option |
| Calibration with the English subject prompt built its content-free inputs with the Sinhala subject name | wrong prior for that condition | content-free inputs copy the real row's metadata |
| **No answer pre-fill; Gemma encodes the space before a digit as a separate token** | label mass 0.05, accuracy close to random | pre-fill "*answer-word*:␣" (Section 5.2) |
| RunPod presets `HF_HOME` on the 30 GB volume | disk full during the MADLAD download | model cache on the 50 GB container disk |
| MADLAD with beam 4 and batch 32 ran out of GPU memory (24 GB) | wave-3 translation failed | automatic batch halving on out-of-memory |

| Problem | Effect | Fix |
|---|---|---|
| Parameter counter counted tied embeddings twice | Gemma 4.97B instead of 4.30B; Gemma+MADLAD wrongly reported as over 8B | count from the safetensors headers (Section 4.2) |

# 10 Reproducibility

**Code.** Package `sinmmlu/`: `data` (loading, normalisation, ids, permutations), `prompts` (prompt registry), `scoring` (averaging, calibration), `model` (loading, label/cloze scoring, generation), `evaluate` (runs conditions), `translate`, `ensemble`, `rationales`, `report` (paired statistics), `select_winner`, `predict_winner`, `submit` (format validation), `build_dev`, `dedup`, `tokens`, `debug`. 34 offline tests (including a fake bigram model that checks batched label and cloze scoring against an unbatched reference, and a simulated out-of-memory split) run without a GPU. Every experiment stores its configuration, model revision, parameter count, library versions, GPU and git revision in `outputs/<experiment>/config.json`.

**Commands** (RunPod, after uploading the code and data):

```
export HF_TOKEN=...                       # read token; Gemma licence accepted
bash runpod/setup.sh                      # GPU check, pinned libraries, dev files, tests, model downloads
bash runpod/run_subset.sh smoke           # then wave1, wave2, wave3, combos, confirm
bash runpod/select_and_test.sh            # selection rule -> winner on the 2,238 test questions
# or directly, the final system only:
python -m sinmmlu.evaluate --model configs/models/gemma3_4b_it.yaml --data test \
       --prompt cot_en_role --chat on
```

The submission is written to

`outputs/gemma3_4b_it__none__cot_en_role__chaton__p1__c0__trnone__test/predictions.zip`

Greedy decoding makes the run deterministic up to floating-point differences between GPUs.

**Cost.** All screening, combination, confirmation and test runs together took about 4.5 GPU-hours on one RTX 3090: a label-scored condition takes 5–42 s for 280 questions (few-shot prompts are the slowest), a reasoning condition 6–9 minutes.

# 11 Limitations

- **All development data is Easy.** We have no labelled Medium or Hard questions, so the gains may not transfer to O/L and A/L questions, which are two thirds of the metric. The lower parse rate on the test set suggests longer reasoning is needed there.
- **Single model, no training.** Fine-tuning (for example on Global-MMLU-si [5] and the public training data) was out of time; the paper rates the naturalness of Global-MMLU-si questions at 71/100 against 97/100 for native questions, so such data would need care.
- **Screening noise.** With 280 questions, gains below about 7 points cannot be separated; two subset results did not replicate. Only the confirmation results should be quoted as findings.
- **Parameter-count error** at selection time (corrected; no effect on the decision).
- **Process note.** When the test-set predictions were checked, the counts of predicted answers (1–4) were printed once. Nothing was chosen from them, and no other test-set statistic was looked at.
- **Licences.** NLLB (used only in screening) is CC-BY-NC 4.0; the final system uses only Gemma (Gemma terms of use).

# 12 Next steps

1. Longer reasoning budget (768 tokens) to raise the test parse rate, checked on the confirmation set first.
2. Fine-tuning with LoRA on Global-MMLU-si (and the public training data, if the team agrees) with the answer-digit loss, judged by the same confirmation protocol.
3. MADLAD translation combined with English reasoning (7.24B parameters, within the limit).

# References


[1] Ashmari Pramodya, Nirasha Nelki, Heshan Shalinda, Chamila Liyanage, Yusuke Sakai, Randil Pushpananda, Ruvan Weerasinghe, Hideteka Kamigaito, and Taro Watanabe. 2025. SinhalaMMLU: A Comprehensive Benchmark for Evaluating Multitask Language Understanding in Sinhala. In *Proceedings of EMNLP 2025*.

[2] Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. 2021. Calibrate Before Use: Improving Few-Shot Performance of Language Models. In *Proceedings of ICML 2021*.

[3] NLLB Team. 2022. No Language Left Behind: Scaling Human-Centered Machine Translation. *arXiv:2207.04672*.

[4] Sneha Kudugunta et al. 2023. MADLAD-400: A Multilingual and Document-Level Large Audited Dataset. In *NeurIPS 2023 Datasets and Benchmarks*.

[5] Shivalika Singh et al. 2025. Global MMLU: Understanding and Addressing Cultural and Linguistic Biases in Multilingual Evaluation. In *Proceedings of ACL 2025*.

[6] Gemma Team. 2025. Gemma 3 Technical Report. *arXiv:2503.19786*.


# A Prompts (verbatim)

Shared pieces (Appendix B of [1]): head = "පහත දැක්වෙන්නේ බහුවරණ ප්‍රශ්නයකි." ("The following is a multiple choice question."); head-subject = "පහත දැක්වෙන්නේ {subject} සම්බන්ධ බහුවරණ ප්‍රශ්නයකි."; instr = "පහත දැක්වෙන ප්‍රශ්නය සඳහා නිවැරදි හෝ වඩාත්ම ගැළපෙන පිළිතුර 1, 2, 3, 4 පිළිතුරුවලින් තෝරන්න." ("Choose the correct or most suitable answer to the following question from the answers 1, 2, 3, 4."); body = "ප්‍රශ්නය: {question} / 1. … / 4. … / පිළිතුර:".

| Name | Text (user turn; "/" = line break) |
|---|---|
| `paper_nosubject` (A0) | head / instr / / body |
| `paper_subject_si` (B1) | head-subject with the Sinhala subject name / instr / / body |
| `paper_subject_en` (B2) | as B1 with the English subject name |
| `strict_nosubject` (B3) | head / instr / "ඔබේ පිළිතුර ලෙස 1, 2, 3 හෝ 4 යන ඉලක්කම පමණක් ලියන්න. වෙනත් කිසිවක් ලියන්න එපා." ("Write only the digit 1, 2, 3 or 4 as your answer. Do not write anything else.") / / body |
| `role_subject_si` (B4) | "ඔබ {subject} විෂය පිළිබඳ ප්‍රවීණ ශ්‍රී ලාංකික ගුරුවරයෙකි." ("You are an expert Sri Lankan teacher of the subject {subject}.") / then B1 |
| `en_instr_nosubject` (B5) | "The following is a multiple choice question in Sinhala. / Choose the correct or most suitable answer from the options 1, 2, 3, 4. / / Question: {Sinhala question} / options / Answer:" |
| `letters_nosubject` (B6) | as A0 with "A, B, C, D" in instr and options labelled A.–D. |
| `fewshot`$k$`_nosubject` (B7) | $k$ earlier turns (A0 text of an example / "පිළිතුර: $d$"), then A0 |
| `fewshot3_subject_si` | 3 earlier turns in B1 format, then B1 |
| `cloze_nosubject` (C4) | "පහත දැක්වෙන්නේ ප්‍රශ්නයකි. නිවැරදි පිළිතුර ලියන්න." ("The following is a question. Write the correct answer.") / / "ප්‍රශ්නය: {question} / පිළිතුර:" (no options) |
| `cot_si` (C6) | head / instr / "පිළිතුර තෝරා ගැනීමට පෙර පියවරෙන් පියවර සිංහලෙන් තර්ක කරන්න. අවසානයේ ``පිළිතුර: <අංකය>'' ලෙස ලියන්න." ("Before choosing the answer, reason step by step in Sinhala. At the end write 'Answer: <number>'.") / / question / options |
| `cot_en` (C7) | "The following is a multiple choice question in Sinhala. / Think step by step in English, then choose the correct or most suitable answer from the options 1, 2, 3, 4. / End your reply with "Answer: <number>". / / Question: {Sinhala question} / options" |
| `cot_en_subject` | as C7 with "…question in Sinhala about {English subject}." |
| `cot_en_role` (final) | "You are an expert Sri Lankan teacher of {English subject}." / then `cot_en_subject` |
| `cot_en_fewshot2` | 2 earlier turns (C7 text of an example / the model's own correct English reasoning), then C7 |

| Name | Text (user turn; "/" = line break) |
|---|---|
| `en_nosubject` (D1–D3) | "The following is a multiple choice question. / Choose the correct or most suitable answer from the options 1, 2, 3, 4. / / Question: {English translation} / translated options / Answer:" |
| `bilingual_nosubject` (D4) | "The following is a multiple choice question in Sinhala, followed by a machine translation into English. / The translation may contain errors; the Sinhala text is the original. / Choose …/ / Sinhala: / ප්‍රශ්නය: … / options / / English: / Question: … / translated options / Answer:" |
| `si_q_en_opts` (D5) | A0 with the options replaced by their English translations |

# B Example of a model-written worked example

Beginning of the English reasoning that Gemma wrote for an Arts pool question (gold answer 4), used as a worked example in `cot_en_fewshot2`:

```
Okay, let's break down this Sinhala multiple-choice question. The question describes a
statue or image of deities (likely Lakshmi and Saraswati) where one leg is bent and the
other leg is extended downwards. We need to identify the correct posture described.
...
Answer: 4
```

# C Prediction distributions on the screening set

| Share of predictions (%) | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Gold answers | 27.5 | 26.4 | 26.4 | 19.6 |
| A0 baseline | 22.9 | 30.0 | 33.6 | 13.6 |
| 4 option orders | 26.1 | 26.4 | 26.1 | 21.4 |
| Calibration | 5.4 | 46.8 | 35.0 | 12.9 |
| Gemma-3-4B-pt (base model) | 46.4 | 35.7 | 11.8 | 6.1 |
| `cot_en` | 25.0 | 26.8 | 29.6 | 18.6 |
| `cot_en_role` | 26.4 | 26.8 | 26.8 | 20.0 |

**Table 13:** Predicted-option shares on the screening subset. Reasoning prompts give the most balanced predictions.

# D Timeline

- 14–15 Oct: student phase: data checks (one label fixed), dedup, phase-0 evaluation of 0.6–1B models on a Colab T4, three scoring bugs found by review.
- 24 Oct: code rebuilt as the `sinmmlu` package; base model fixed to Gemma-3-4B-it; screening subset built; RunPod RTX 3090 set up; smoke test (pre-fill bug found and fixed); waves 1–3; combinations; confirmation; first test predictions.
- 27 Oct: leaderboard closes. 28 Oct: code. 30 Oct: report and slides.